

The endocrine system

The endocrine system consists of glands which produce (secrete) chemicals called **hormones**. The human endocrine system is illustrated in Figure 2.

An endocrine gland secretes its hormones directly into the blood which transports them around the body. In contrast a non-endocrine gland, like the pancreas, secretes its products through a tube called a **duct** which leads to the area where these products are used. Since endocrine glands have no ducts they are called **ductless glands**. After entering the blood a hormone travels to all parts of the body but only parts called **target organs** respond to them.

Hormones are produced in minute quantities but their effects on the body are often profound. Some of these effects are temporary. Adrenalin for example quickens the heart-beat, releases sugar from the liver, etc., but these changes last only a few minutes. Some hormones from the pituitary gland, however, have long-lasting effects. They influence the size of the body and mental ability.

The endocrine and nervous systems are both concerned with controlling organs and tissues so that they function together as a co-ordinated **organism** rather than as a collection of independent parts. Control of this type is called **integration**. The endocrine and nervous systems also enable an organism to respond to temporary and long-lasting changes in its surroundings. This is called **adaptation**. There are, however, a number of fundamental differences in the way that hormones and nerve impulses integrate the body and adapt it to its surroundings.

Differences between hormonal and nervous co-ordination

Nerve impulses travel at great speed along nerve fibres whereas hormones travel more slowly in the bloodstream. Nerve impulses travel along a nerve to only one part of the body but hormones are transported to all parts of the body and affect several target organs. When nerve impulses reach a muscle, for example, they produce an immediate response lasting only a few seconds. Hormones produce slower responses which can last for years.

Control of endocrine glands

A few endocrine glands control themselves, but the majority are controlled by the pituitary gland. Therefore the pituitary is called the 'master gland' of the endocrine system.

A good example of how the pituitary controls other endocrine glands is the way that it regulates the flow of the hormone thyroxine from the thyroid gland. The pituitary gland is very sensitive to the amount of thyroxine in the blood. Whenever thyroxine falls below a certain level the pituitary produces a substance called **thyroid-stimulating hormone (TSH)**. This stimulates the thyroid to produce more thyroxine. When thyroxine rises above a certain level the pituitary produces less TSH and so less thyroxine is produced by the thyroid. Constant repetition of these events controls thyroxine production.

This mechanism is an example of **negative feedback**: any change in the level of thyroxine produces an opposite (negative) response which reverses the change. Information about changes in the thyroxine level is constantly 'fed back' to the pituitary so that it can make an appropriate response (Fig. 1). The pituitary controls several other endocrine glands in the same way.

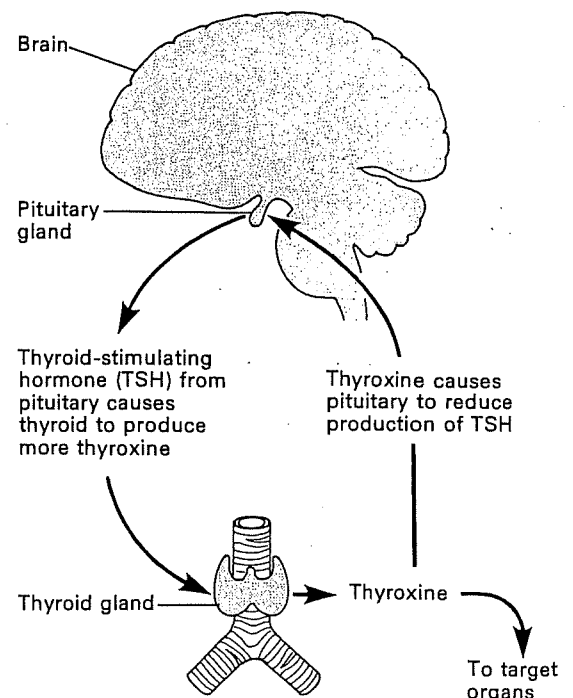


Fig. 1. The pituitary gland controls most other endocrine glands. For example, whenever thyroxine production decreases, the pituitary releases thyroid-stimulating hormone (TSH); but when thyroxine production increases the pituitary produces less TSH.

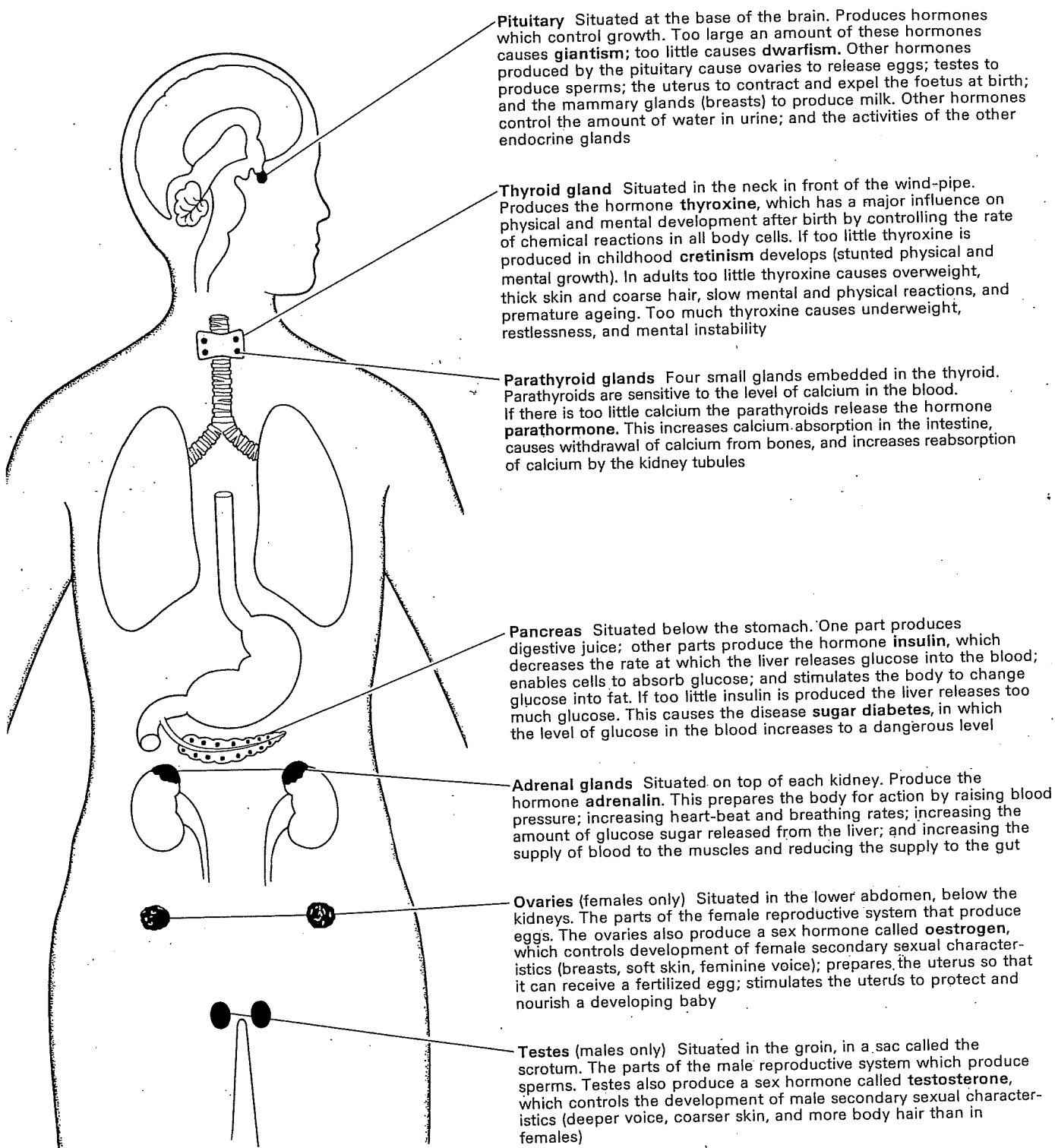


Fig. 2. Human endocrine glands and their functions.

Chemical control

There are small chemical messengers in your body that are responsible for big changes. Hormones control many functions of your body. Hormones are responsible for the changes of puberty as well as being responsible for controlling water balance and the volume of glucose in your blood. The glands of the endocrine system release hormones.

The endocrine system

The endocrine system is a communication system that controls the internal environment of the body. Water and glucose levels in your blood, body temperature and the changes the body goes through during puberty and pregnancy are all controlled by the endocrine system.

Hormones are chemical substances that act as messengers in the body. They are produced by the endocrine glands, which are located all around your body as seen in Figure 7.3.1. Together, all these glands form the endocrine system.

Pituitary gland: produces the hormones:

- human growth hormone (HGH), which controls cell growth and development
- antidiuretic hormone (ADH), which controls water balance.

Also stimulates other glands to release their hormones.

Pancreas: produces the hormones insulin and glucagon which both control blood glucose levels.

Adrenal: produces the hormone adrenalin in readiness for flight or fight.

Ovaries (in females): produce the hormones:

- oestrogen, which controls female sexual development and the menstrual cycle
- progesterone, which controls the ovary and uterus in pregnancy.

Testes (in males): produce the hormone testosterone which controls male sexual development and sexual activity.

Thyroid: produces the hormone thyroxin which controls the rate of chemical reactions in cells.

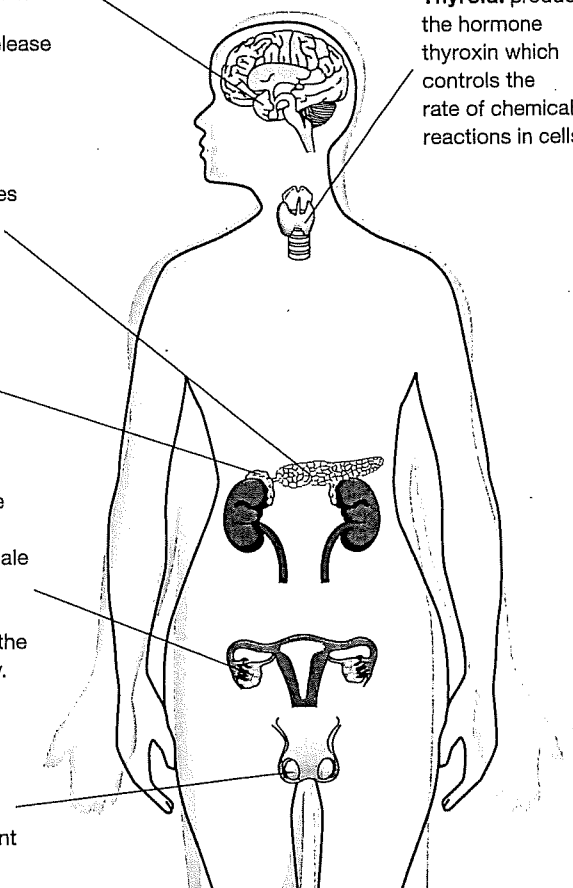


FIGURE 7.3.1 Major human endocrine glands

Chemical messengers

Hormones are secreted in very small quantities and travel through the bloodstream to all parts of the body. However, only particular cells called **target cells** will respond to particular hormones. Other cells don't have the particular receptor needed to read the message being sent. Hormones have a specific chemical structure and specific shape that fits chemically onto a receptor on the cell membrane of the target cell—just like pieces of a jigsaw puzzle. In this way they act very similar way to the lock-and-key model found in enzymes. In this way hormones target the correct type of cell without others being affected. When the hormone binds to the receptor on the target cell, this will start changes in the activities of the cell. This process is shown in Figure 7.3.2.

Hormones are only needed in very small quantities to have an effect on the body. They regulate functions like growth, water balance, regulation of glucose and sexual development.

Systems working together

The endocrine system is coordinated by the **pituitary gland**, which responds to information from the **hypothalamus**. The **hypothalamus** is a portion of the brain. You can see its location in Figure 7.3.3. It constantly checks the internal environment—that is, the conditions within the tissues, organs and systems of your body. If these conditions change, then the hypothalamus responds.

The most important function of the hypothalamus is to link the nervous system and the endocrine system. It secretes (releases) hormones that act on the pituitary gland.

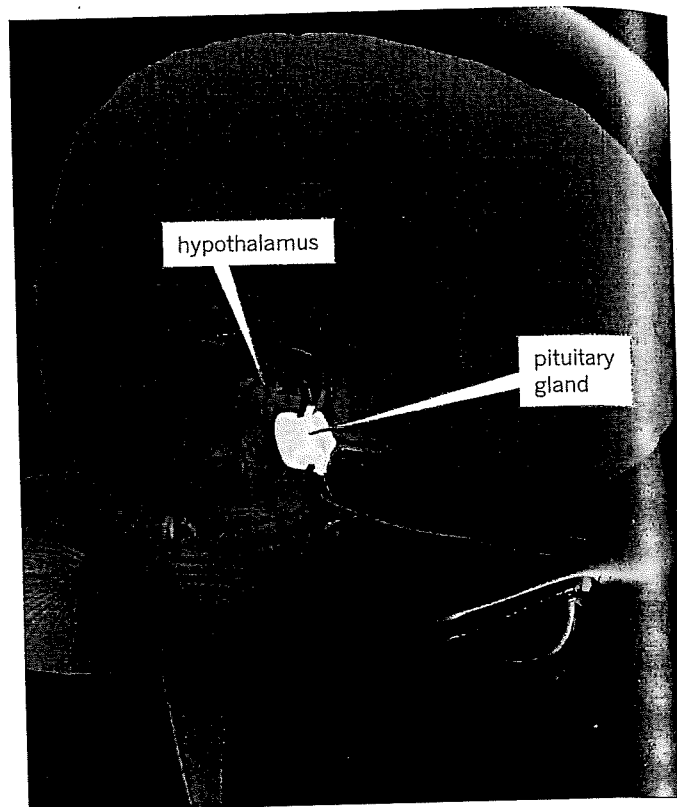


FIGURE 7.3.3 Two parts of the endocrine system: the hypothalamus, and the pituitary gland that attaches to it. These are located deep within the skull, where they are well protected.

The **pituitary gland** is often called the 'master gland' because it controls the activities of other endocrine glands such as the ovaries, the testes and the thyroid gland. The pituitary gland responds by hormones coming from the hypothalamus secreting other hormones or producing less of the hormones. Through its action on the pituitary gland, the hypothalamus controls important aspects of the body such as body temperature, rate of metabolism and water content.

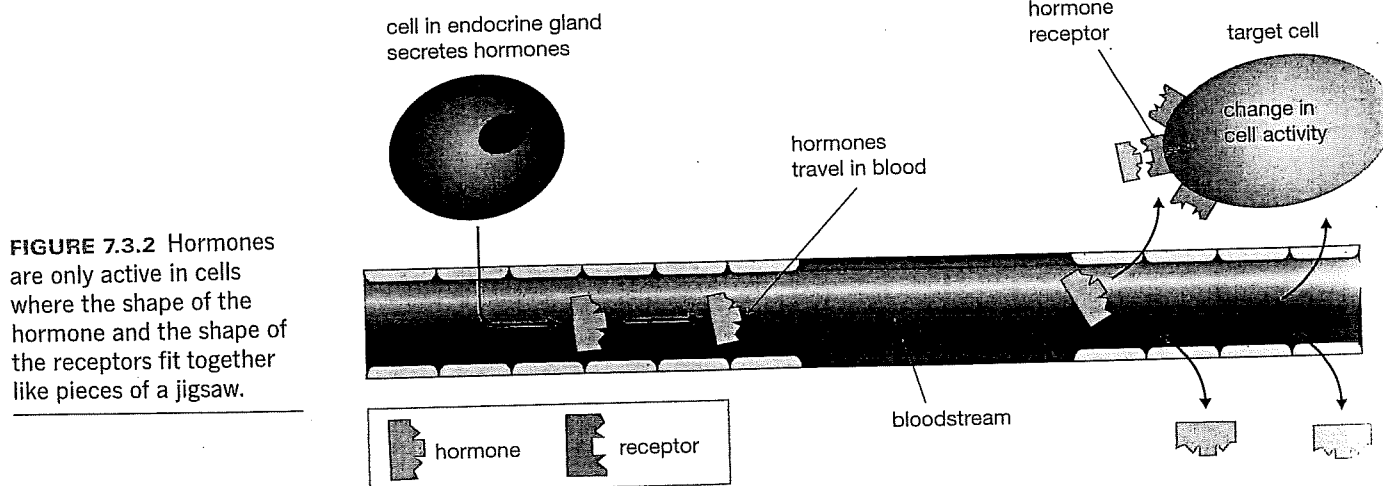


FIGURE 7.3.2 Hormones are only active in cells where the shape of the hormone and the shape of the receptors fit together like pieces of a jigsaw.

In situations of fear or stress both the nervous system and endocrine system have roles to play (Figure 7.3.4). This response is known as the 'fight or flight response'. The endocrine system releases the hormones adrenaline and cortisol, which increases heart rate, breathing rate and blood pressure. Blood glucose levels also rise to provide muscles with required energy for this response. The nervous system acts swiftly through the network of neurons to work with the endocrine system to increase the rate of breathing and increase the heart rate. Pupils will dilate (widen) to improve vision, sweat glands will produce more sweat and digestion will slow down or cease.



FIGURE 7.3.4 The actions of the nervous system and the endocrine system cause the sensations associated with fear or stress.

Controlling the internal environment

Your body works most efficiently when its internal environment is kept constant. This means that factors such as temperature, water content, available energy, available oxygen and concentration of wastes in the blood are all controlled. The process of maintaining a constant internal environment is known as **homeostasis** (*homeo* means 'same' and *stasis* means 'state').

Homeostasis involves receptors that are sensitive to a particular stimulus, and effectors, muscles or glands that have an effect on the same stimulus.

This type of control is known as a **feedback system**.

To understand this, consider a reverse-cycle air conditioner like the one shown in Figure 7.3.5 as an example of a machine that maintains a constant environment. A sensor called a thermostat is set at a particular temperature range, such as 21–23°C.

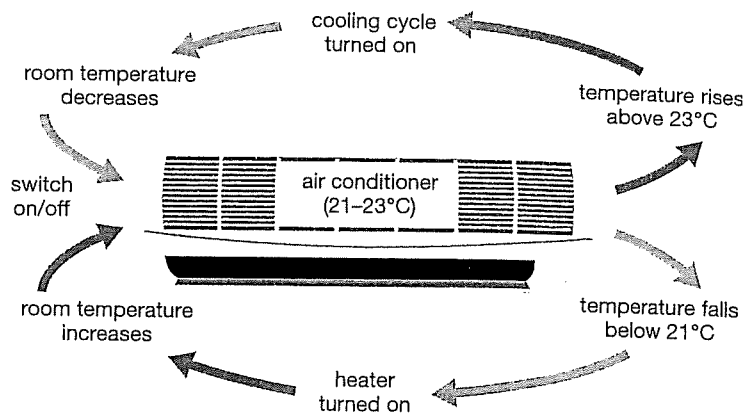


FIGURE 7.3.5 A reverse-cycle air conditioner maintains a preset temperature in a room by heating or cooling the air in response to temperature detectors.

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If the temperature of the room goes above 23°C, the air conditioner switches on and cools the room until the required temperature is reached. The sensor detects this and then the air conditioner switches off until it once more detects a rise in temperature. If the sensor detects a lower temperature than 21°C, the heating system turns on. The temperature in the room rises and the heater turns off once the set temperature range of 21–23°C is reached.

Your body responds to the environment in a similar way. When temperatures rise, the body reacts by sweating and dilating blood vessels in an attempt to cool a person down. When the temperature decreases, the body responds by contracting blood vessels, creating goosebumps and making hairs stand up on end in an attempt to insulate the body and prevent heat loss. In this way, the body maintains its optimal temperature.

Controlling body temperature

Digestion, growth and repair, respiration and manufacture of hormones are some of the chemical reactions taking place inside your body. The heat they produce as a by-product maintains your body temperature regardless of the temperature of your surroundings. This is illustrated in Figure 7.3.6 on page 296. Because you can maintain a constant body temperature, you are said to be **endothermic**. However, if the temperature inside your body was to increase by more than a few degrees above 37°C, your metabolism would stop, and you would die. If your body temperature fell below 37°C, your metabolism would slow down.

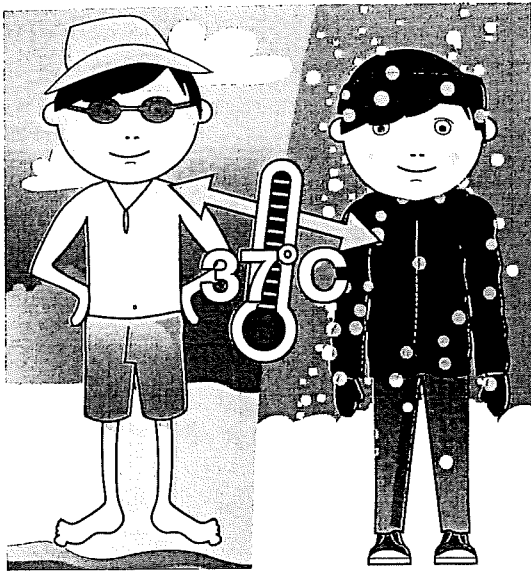


FIGURE 7.3.6 Your body temperature stays the same regardless of the temperature of the environment. There are things you can do to help your body, such as adding clothing when it is cold, and removing clothing and using shade when it is very hot.

One of the most important examples of homeostasis is the regulation of body temperature. Not all animals can do this as a normal body function. Animals that can maintain a constant internal body temperature such as birds and mammals are called **endothermic**. Endothermic animals are sometimes called warm-blooded animals because they maintain their body temperature within narrow limits. These narrow limits are controlled by the nervous and endocrine system; they maintain an internal body temperature between 36 and 40°C. Animals that cannot maintain a constant internal body temperature such as reptiles are called **ectothermic**. Figure 7.3.7 demonstrates the relationship between body temperature and environmental temperature for ectothermic and endothermic animals.

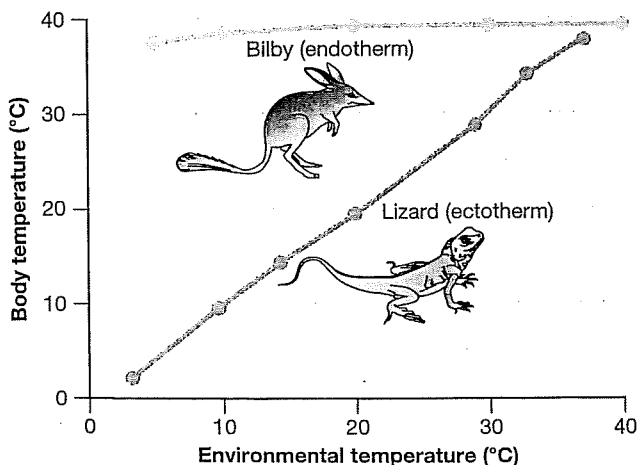


FIGURE 7.3.7 Endothermic animals can maintain their body temperature in a changing environment. Ectothermic animals have a body temperature similar to that of their surroundings.

Ectothermic animals can also have very warm blood; however, they regulate their body temperature through behavioural mechanisms such as by lying in the sun when cold and moving into the shade when hot, as seen in Figure 7.3.8. Some other ectotherms such as bees and tuna use extended muscular activity to increase body temperature. These behavioural mechanisms of controlling body temperature can be very effective especially when combined with internal mechanisms that ensure that the temperature of the blood going to vital organs such as the brain and heart is kept constant.

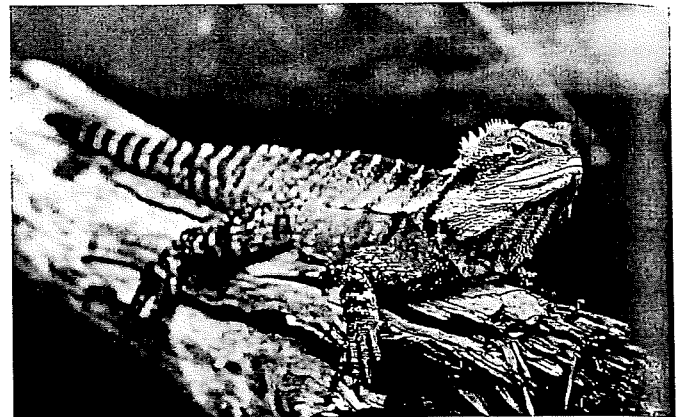


FIGURE 7.3.8 Like all lizards, the Australian water dragon is ectothermic. This dragon has flattened itself out to maximise the amount of sunlight falling on it.

Hormonal control of temperature

The hypothalamus acts on the pituitary gland to control body temperature through the action on another endocrine gland—the thyroid gland.

The hypothalamus receives information from temperature receptors in the skin as shown in Figure 7.3.9 and from internal receptors, including the hypothalamus itself.

If the hypothalamus detects a fall in body temperature, it produces a hormone that causes the pituitary gland to secrete more thyroid-stimulating hormone (TSH). TSH stimulates the thyroid to release more of the hormone thyroxine. Thyroxine travels in the blood to all cells and causes the rate of metabolism in the cells to increase.

Increased metabolism generates more heat and warms the body. Producing the hormones that cause these changes takes time, and therefore the endocrine system does not have immediate control over body temperature.

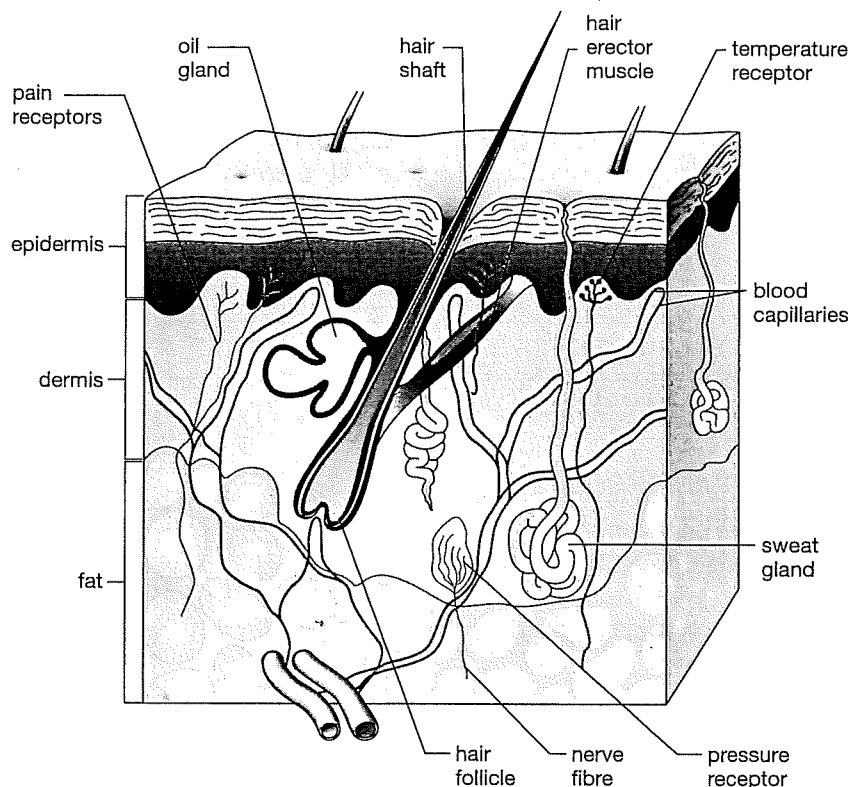


FIGURE 7.3.9 The skin has many receptors and provides you with a lot of information about your surroundings.

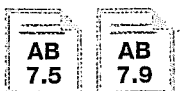
Nervous control of temperature

Body temperature is also controlled by the nervous system and this is a more immediate response. When the hypothalamus detects a drop in temperature, it sends nerve impulses to muscle groups around vital organs such as the heart and lungs. Small shaking movements begin in these muscles. Eventually, the shaking movements extend to the large muscles of the arms and legs, and you begin to shiver. Shivering increases the activity of muscle cells, producing heat and raising body temperature. This is the body's way of creating warmth by using energy.

Another aspect of nervous control is the process that reduces blood flow to your skin when you are cold. The sympathetic nervous system causes a narrowing of the blood vessels near the surface of the skin. This reduces blood flow and therefore heat flow to the skin. If the external temperature is very low, the blood flow to the fingers, toes, nose and ears is reduced further and you can lose feeling in them. This causes your toes and fingers to go numb.

When the hypothalamus detects a rise in body temperature, nerve messages are sent to the sweat glands and blood vessels. Blood vessels close to the skin dilate (increase in diameter). This change allows more blood and the heat it carries to reach the skin surface. The extra blood near the surface makes your skin red.

The message from the hypothalamus causes the sweat glands to produce more sweat. Heat from your body causes the sweat to evaporate. The rate at which heat is lost by evaporation depends on the difference in temperature between the body and the surrounding air, and the relative humidity of the air.



Review questions LS

Remembering

- Define the terms: (4 marks)
 - hormone
 - endothermic
 - effector
 - receptor.
- What term best describes each of the following? (4 marks)
 - unable to maintain a constant internal body temperature
 - the type of cells that respond to hormones
 - the process of maintaining a relatively constant internal environment in the human body
 - the gland in the body referred to as the 'master gland'.
- What three functions in your body are regulated by hormones? (3 marks)
- Refer to Figure 7.3.9 on page 297 and list three types of receptors found in the skin.

Understanding

- Describe the relationship between hormones and the endocrine system. (2 marks)
- What is the function of the hypothalamus? (1)
 - Why can the hypothalamus be thought of as part of both the nervous system and the endocrine system? (2 marks)
- Explain why sweating is an efficient way for the body to lose heat. (2 marks)
- Why is it important that your body temperature remains constant? (2 marks)
 - Describe two involuntary reactions that keep your body temperature from rising. (4 marks)
 - Describe one involuntary reaction that prevents your body temperature from falling. (2)
- How are hormones transported in the body? (2)
- The body makes more than 20 hormones, each with a specific function. Outline the process by which these hormones recognise which tissues and organs to communicate with. (4 marks)

Applying

- Identify the endocrine glands labelled A–G in Figure 7.3.10. (7 marks)

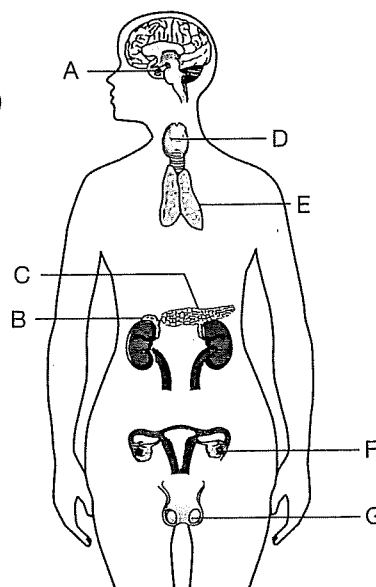


FIGURE 7.3.10

- Use a diagram to help you explain how hormones recognise their target cells. (4 marks)

Analysing

- Use examples to compare the way that endothermic and ectothermic animals control body temperature. (4 marks)
- Analyse why:
 - ectothermic animals generally live in warmer climates (2)
 - endothermic animals can live in a wide range of environments, from hot to cold. (2)

Evaluating

- Explain why the pituitary gland is often referred to as the 'master gland'. (4)

Create

- optional
- Create a model or digital animation to demonstrate the action of hormones. Begin with the hormones being produced in the gland and end with the target cells responding.
 - Create a diagram to represent the response your body would have to an increase in environmental temperature.

Practical investigations



SPARKlab alternative
available for this activity.

1 • Changing temperature

Processing
& Analysing

Evaluating

Purpose

To investigate the effect of exercise on body temperature.

Hypothesis

What do you think happens to your skin and internal body temperatures as you exercise? Before you go any further with this investigation, write a hypothesis in your workbook.

Timing 60 minutes

Materials

- stopwatch
- electronic clinical thermometer (if available, data-logging equipment could be used for this experiment)
- area of the school where you can run around

Students with health problems may not be able to take part in this activity.

Procedure

- 1 Work in pairs. One person is the subject and the other is the recorder.
- 2 Record the skin temperature of the subject at the start of the experiment. Measure the skin temperature by holding the thermometer inside a bent elbow. Measure the internal temperature at the ear using an electronic clinical thermometer.
- 3 The subject undertakes 10 minutes of vigorous exercise, enough for them to feel hot and possibly turn red in the face.
- 4 The recorder records the subject's skin temperature and internal temperature.
- 5 Observe and record any other changes resulting from the exercise in an appropriate table.

Results

Record all measurements and observations.

Review

- 1 Describe the changes to the skin temperature during the experiment.
- 2 Describe the changes to the internal body temperature during the experiment.
- 3 Compare the changes in skin temperature with the changes in internal body temperature.
- 4 Describe any other changes that were observed.
- 5 How could these other changes contribute to the changes you saw in temperature?
- 6 a Construct a conclusion for your investigation.
b Assess whether your hypothesis was supported or not.



Variation in body temperature

Variation in body temperature can arise due to a number of factors;

- External influences eg hot & cold surroundings
- Rate of activity eg exercise/work rate
- Biorhythms – 24hr (↓morning ↑afternoon)
- Menstrual cycle (↑2nd half due to luteinising hormone and progesterone)

As a general rule the body needs to balance heat gain and heat loss. Heat is gained from body processes such as cell metabolism and muscular exercise, and from conduction and radiation from the environment. Heat is lost from the body by conduction, convection, radiation and evaporation.

You need to understand the mechanism and relative importance of each in gaining and losing heat.

Cell metabolism is the sum of all the chemical reactions that occur in the body to keep humans alive. The majority of these chemical reactions produce 'waste' heat which is used to maintain our core temperature.

Muscular exercise results in large amounts of heat being produced as by product of the contraction of muscles.

Conduction is a type of heat transfer where heat energy moves from a warmer object to a cooler object when they are in direct physical contact. If you touch an object that is cooler than your body, heat passes from your body to the cooler object. The reverse happens when you touch an object that is hotter.

Convection is a type of heat transfer where heat energy is transferred by the movement of fluids. Warm air (created by contact with a warm body) rises and is replaced by cooler air. The currents of moving air remove heat energy from the body.

Radiation is a type of heat transfer where heat energy moves from a warmer object to a cooler object across space, no contact between the bodies is necessary. If your body is at a higher temperature than the environment it will radiate heat to the environment cooling the body. If your body is cooler than the environment then it will gain heat from the environment through radiation.

Evaporation occurs when liquid water is converted to water vapour. This requires energy, and this energy is taken from the body in the form of heat, thus cooling the body. When humans perspire or sweat the evaporation of the liquid from the skin cools the body. In general humans are faced with one of two problems, the need to gain more heat due to too low a core temperature or to lose heat if the core temperature rises over 37° C.

Regulation of Body Temperature

Humans are termed 'homeotherms' that is we can maintain a fairly constant body temperature independent of the external temperature in most environments. Humans have a 'thermostat' within the brain that measures the temperature of the blood and attempts to maintain it around 37° Celsius. This is known as the core temperature, other parts of the body especially the extremities can be at a higher or lower temperature for short periods of time.

The advantage of a constant body temperature is that the metabolic chemical reactions that occur in our cells can be tailored to work at their optimum rate. This is done by producing enzymes that assist the reactions by working at their best when the surrounding temperature is 37° C.

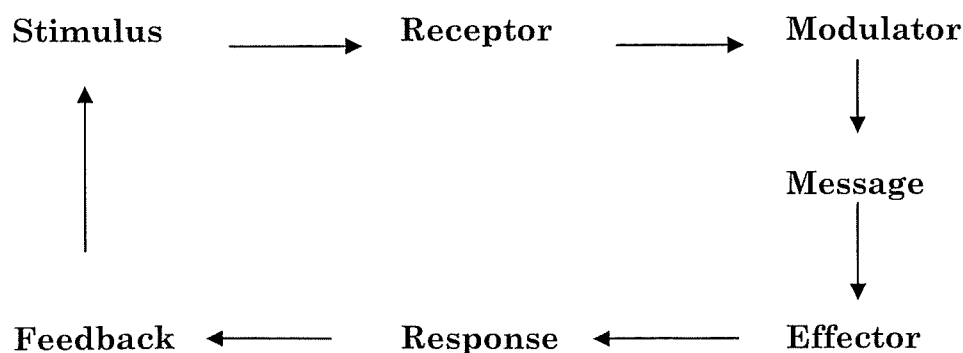
The disadvantage of being a homeotherm with a constant body temperature is that problems occur when that temperature is not maintained. Too high or low temperatures can result in the denaturation of enzymes, nerve malfunction and even death. As a general rule the body can only survive for a short period of time if the core temperature gets too high or low. A 5 degree increase or decrease can lead to brain damage and a 7 degree differential can result in death.

The stimulus response model

Homeostasis is the name given to the concept of maintaining a constant optimal environment irrespective of the external one.

Control mechanisms function at the level of cells, physiological systems and behaviour, and compensate for changes in the external or internal environment. To maintain homeostasis the body requires control mechanisms that operate automatically. These mechanisms usually occur as steady state models.

Steady state models are usually presented as follows.

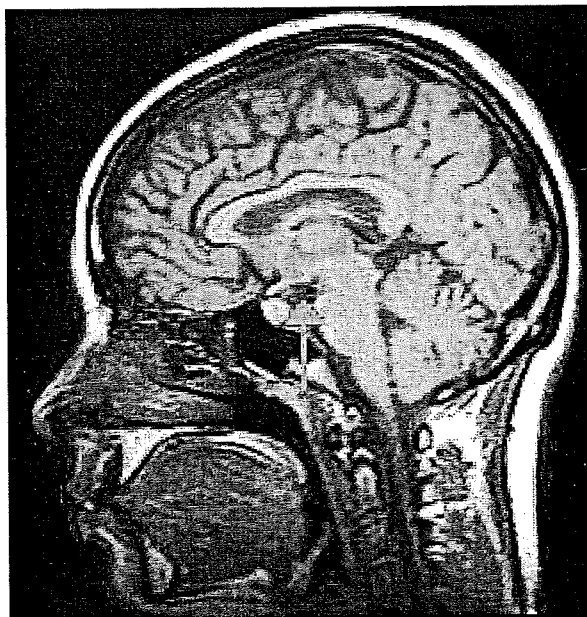


The stimulus is the factor that changes and causes the system to operate, the receptor detects the change in the stimulus; the modulator processes information from the receptor and sends an electrical (nervous) or chemical (hormonal) message to the effector. The effector carries out the response, the response is the change in the effector to maintain homeostasis and the feedback is how the response modifies the stimulus.

Preventing body temperature from decreasing

Special sensors in the skin and brain stem called thermo-receptors measure the temperature of the skin and blood and inform the brain that the body is getting cold. The brain then initiates a number of actions;

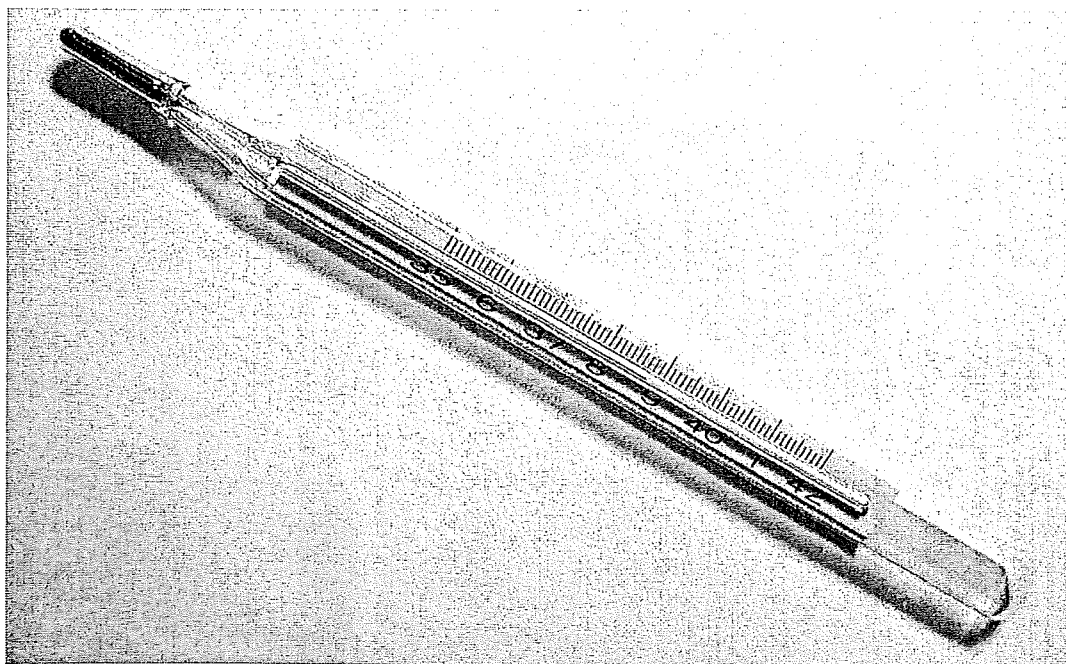
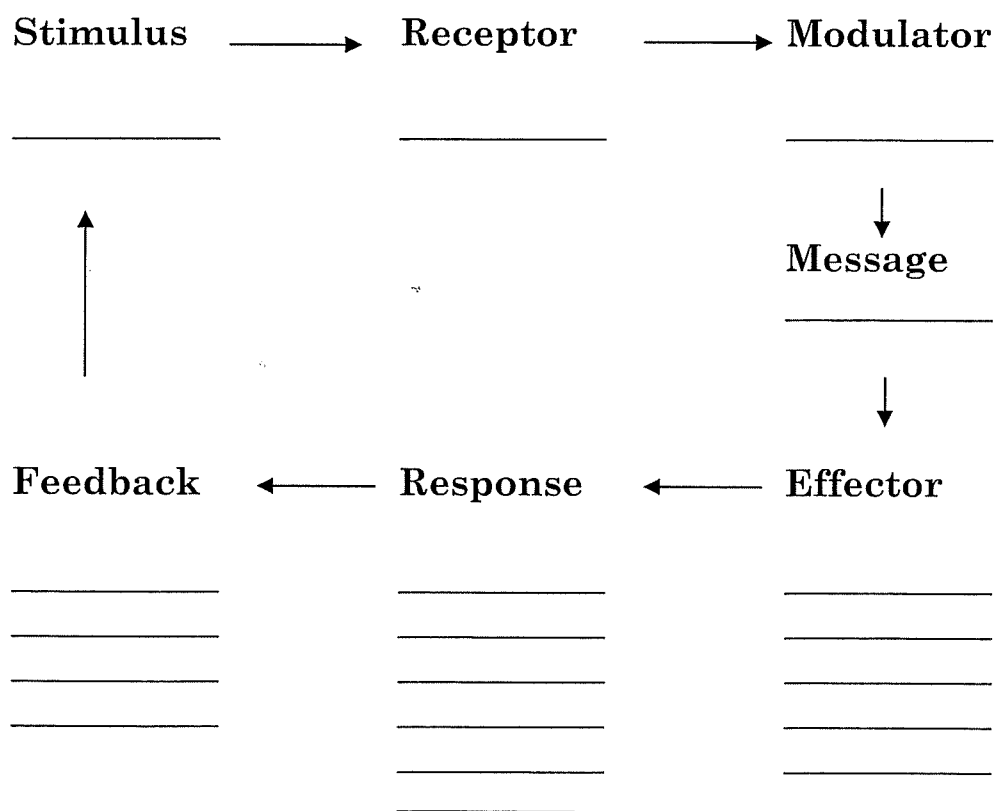
- The blood vessels in the skin narrow in diameter so less warm blood flows to the skin; this is called Vasoconstriction. The decrease in blood to the skin means that less heat is lost by radiation. An observable characteristic of vasoconstriction is an increase in paleness of the skin.
- As a short term measure the voluntary skeletal muscles start a spasmodic reflex contraction. This results in increased heat production. If you can remember being in swimming lessons when you were young you might remember the shivering in your jaw muscles. The heat produce in the jaw muscles helps warm the blood moving up the neck into the brain.
- In the medium term the Thyroid gland produces a hormone called Thyroxine that helps increase the rate of metabolism in all cells therefore producing extra heat.
- The skin reduces/stops sweating so that little evaporation of sweat and subsequent heat loss can occur.
- The part of the brain which initiates changes in behaviour (Cerebral Cortex) causes heat seeking behaviour eg drinking hot liquids, standing near a fire/heater and voluntary activity eg jogging on the spot. It also promotes behaviour that reduces heat loss eg decreasing surface area by huddling with others or making your body a ball shape and putting on more clothing.
- Although not effective in reducing heat loss (as humans have little hair) the hair erector muscles contract to raise the hairs resulting in what is called 'Goose Bumps'.



Use the information above to complete the following stimulus response model below.
Teachers see Teachers Guide

Notes

Preventing body temperature from decreasing



A clinical thermometer

Preventing body temperature from increasing

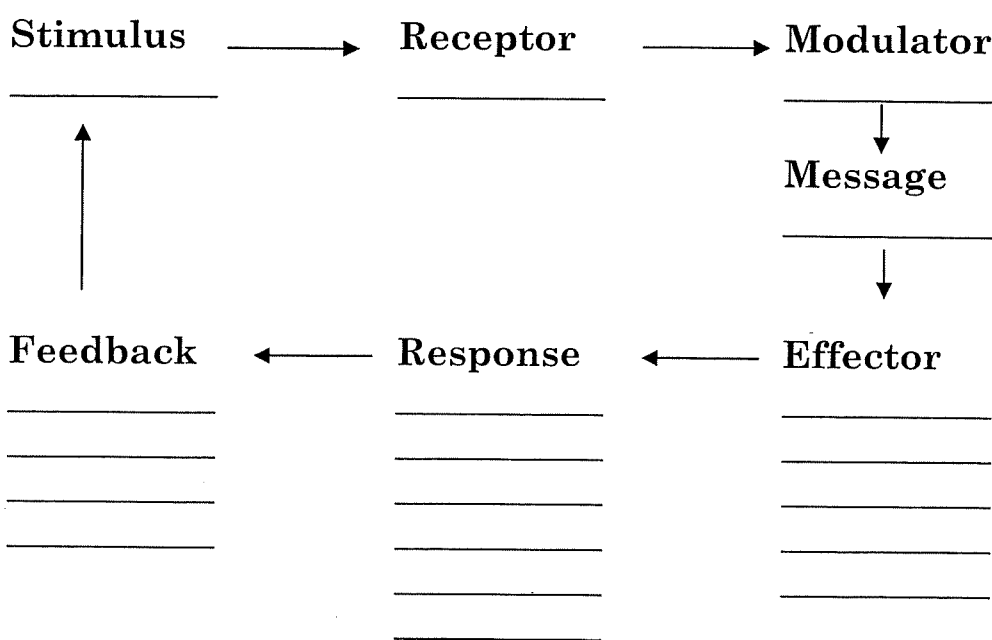
Special sensors in the skin and brain stem called thermo-receptors measure the temperature of the skin and blood and inform the brain that the body is getting too hot. The brain then initiates a number of actions;

- The blood vessels in the skin increase in diameter so more warm blood flows to the skin, this is called Vasodilation. The increase in blood to the skin means that more heat is lost by radiation. An observable characteristic of vasodilation is an increase in redness of the skin.
- In the medium term the Thyroid gland produces less of the hormone called Thyroxine. This helps decrease the rate of metabolism in all cells, producing less heat.
- The skin starts/increases sweating. The evaporation of the sweat from the surface of the skin removes heat from the skin surface. The effectiveness of sweating is largely dependent on the relative humidity level of the air and air movement.
- The part of the brain which initiates changes in behaviour (Cerebral Cortex) causes heat losing behaviour eg drinking cold liquids, standing near a fan/air conditioner, swimming, increasing surface area by standing with arms out and legs spread and reducing the amount of clothing. To help produce less heat within the body it also voluntarily reduces activity eg staying still and not moving around.
- Muscle fibres inside muscles take turns contracting while the muscle is stationary, this contributes to muscle tone. In hot conditions to reduce heat production the fibres will not contract or contract less often resulting in a 'floppy' appearance to muscles, and people feeling 'hot and limp'.

Use the information above to complete the following stimulus response model below.
Teachers see Teachers Guide

Notes

Preventing body temperature from increasing



Library research – dysfunctions

Research and report on the cause, symptoms, effect on the body and treatment for the following dysfunctions. The table has been provided for note taking.

Dysfunctions	Cause	Symptoms	Effect on the body	Treatment
Heat cramp				
Heat stroke				
Heat exhaustion				
Hypothermia				
Chilblains				
Frostbite				

Skin surface temperature

The body has two basic sets of physiological (functional) mechanisms to maintain our body's core temperature. These mechanisms can increase heat loss and reduce heat gain when the body is too hot and decrease heat loss and increase heat gain when it is too cold.

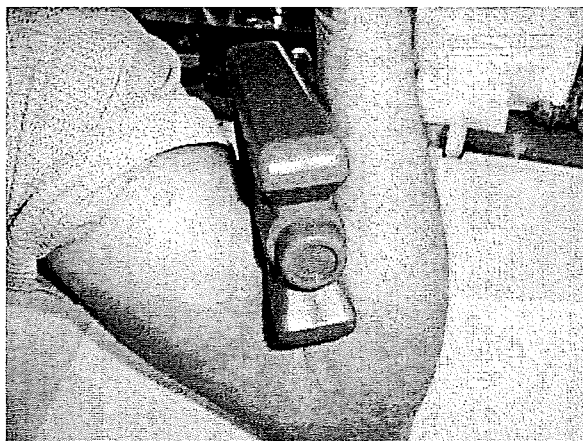
Two of these mechanisms are vasoconstriction and vasodilation. Vasoconstriction reduces blood flow to the skin thus reducing the skin temperature; this reduces the difference between the skin temperature and the environment and slows heat loss. Vasodilation increases blood flow to the skin thus increasing the skin temperature; this increases the difference between the skin temperature and the environment and increases heat loss.

This activity examines the time taken for the skin to return to normal temperature after being exposed to both hot and cold temperatures.

Hypothesis – That a student's skin temperature will return to its normal level quicker after having a cold source applied when compared to a hot source.

Materials

- Frozen cool pack or ice in a plastic bag
- Hot heat pack
- Thermometer (medical)



Procedure

- 1) Place the bulb of the thermometer in the crook of your arm (elbow pit).
- 2) Bend your forearm to hold the thermometer in place.
- 3) Wait for about 1 – 2 minutes until the temperature stabilises, read the temperature and record in the results table.
- 4) Remove the thermometer and apply the cold pack to the crook of your arm, bend your forearm to hold the cool pack in place for approximately 1 minute and then remove the cold pack.
- 5) Place the bulb of the thermometer in the crook of your arm (elbow pit) and bend your forearm to hold the thermometer in place.
- 6) Wait for 1 minute then read the temperature and record in the results table.
- 7) Take a reading every 30 seconds and record the temperature.
- 8) Continue taking readings until the temperature returns and is maintained at the same level as that from step 3 above.
- 9) Repeat steps 4 – 8, using the hot pack.

Results

Time (min:sec)	Temperature after application of	
	Cold pack	Hot pack
Initial/normal		
1:00		
1:30		
2:00		
2:30		
3:00		
3:30		
4:00		
4:30		
5:00		
5:30		
6:00		
6:30		
7:00		

Data analysis

Draw a graph of your data on the grid on the following page, break the axis (if required) and use a scale to suit the data.

Questions

- Examine the data and graph; did the skin temperature return to normal faster after being cooled or being heated?

- Explain some possible reasons for the difference.

- Compare your results to others in the class, did all subjects respond the same way?

- Do you think you would respond in the same way in all areas of your body eg behind your knee?

- How could this experiment be improved to make the data more reliable and valid?

Endocrine glands and hormones

Like the nervous system, the endocrine system co-ordinates and regulates many body processes. The nervous system exerts control by the transmission of nerve impulses to and from the various tissues. The endocrine system influences the activity of cells by release of substances called hormones. In this chapter we will discuss the activities of some of the major endocrine glands. Later chapters, dealing with specific body processes, will consider the action of some hormones in more detail.

Endocrine glands

There are two kinds of glands in the body:

- **Exocrine glands** secrete into a duct that then carries the secretion to the body surface or to one of the body cavities. Sweat glands, mucous glands, salivary glands and the glands of the alimentary canal are examples of exocrine glands.
- **Endocrine glands** secrete into the extracellular fluid that surrounds the cells making up the gland. The secretion then usually passes into the capillaries to be transported by the blood. Endocrine glands are sometimes called ductless glands. The major endocrine glands are shown in Figure 24.1.

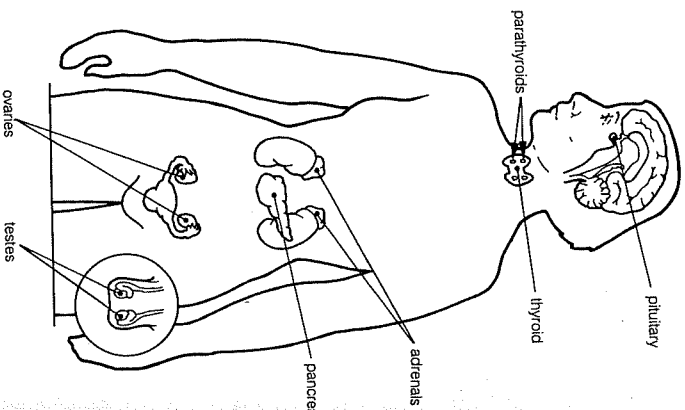


Fig. 24.1 Major endocrine glands

Hormones
The secretion of an endocrine gland is called a hormone. Hormones may be proteins, steroids or amines. They maintain homeostasis by changing the activity of cells, either by changing the permeability of the cell membrane, or by changing the rate of chemical reactions within

the cells. The bloodstream transports most hormones throughout the body. A hormone may affect all the cells of the body or may only affect particular groups of cells, target cells, or particular organs, target organs.
Hormones seem to influence their target cells in two different ways. Protein and amine

hormones attach to receptor molecules on the membrane of the target cell. Different receptor molecules are involved for each hormone so that a hormone will only influence a cell with the correct receptor site. Combination of the hormone with the receptor causes a secondary messenger substance to diffuse through the cell and activate particular enzymes.

Steroid hormones enter target cells and combine with a receptor protein inside the cell. The complex of hormone and receptor then diffuses into the cell nucleus where it activates the genes controlling the formation of particular enzymes. In this way the amount of an enzyme produced by a cell is regulated.

Hormones are not enzymes, but in many cases they exert their influence by changing the activity of enzymes or by changing the concentration of enzymes.

Feedback control of hormone secretion

To maintain homeostasis, the amount of hormone produced by an endocrine tissue must be very closely regulated. Any oversecretion or undersecretion of a hormone will cause the body to function abnormally.

Hormonal secretions are generally regulated by negative feedback systems (see Chapter 20 for a full discussion of how negative feedback operates). The response produced by the secretion of the hormone is opposite to the stimulus that caused the secretion.

Some negative feedback systems involve the nervous system through the release of regulating factors from the hypothalamus of the brain. These factors regulate the function of the pituitary gland. The hypothalamus can secrete releasing factors, which stimulate release of a hormone, or inhibiting factors, which slow down secretion of a hormone. Later in this chapter there is further discussion of how release of a hormone from the pituitary gland is regulated by the hypothalamus.

The pituitary gland

The pituitary gland, or hypophysis, lies just under the hypothalamus and is joined to the hypothalamus by a stalk called the infundibulum. It is not much bigger than a large pea, about 13 millimetres in diameter, but it is absolutely vital to the normal functioning of the body.

Many pituitary hormones regulate the activity of other endocrine glands and for this reason the pituitary is sometimes referred to as the master

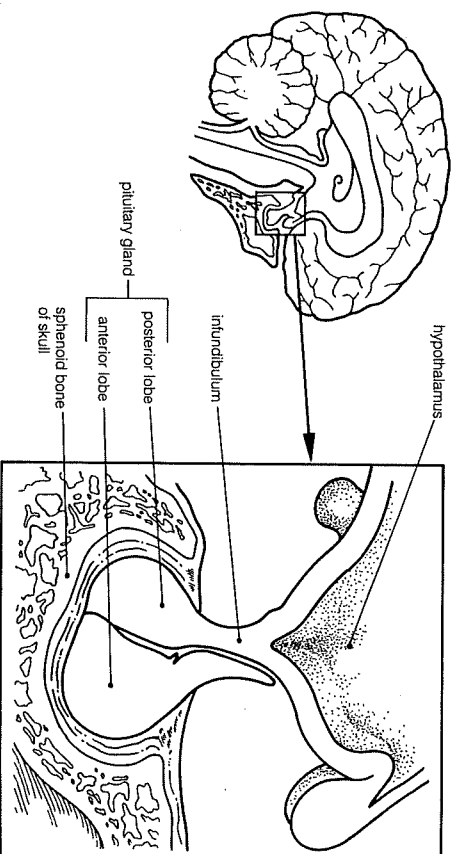


Fig. 24.2 Location of the pituitary gland

gland'. The pituitary consists of an anterior lobe and a posterior lobe, each of which functions separately.

Anterior lobe of the pituitary

The anterior lobe of the pituitary (adenohypophysis) produces a number of hormones that regulate a great range of bodily activities. Secretions of the anterior lobe are in turn controlled by releasing and inhibiting factors secreted by the hypothalamus. These factors are themselves hormones since they are secreted into the extracellular fluid around the cells of the hypothalamus and are carried by the blood to the anterior lobe of the pituitary.

Hormones released by the anterior lobe of the pituitary include the following:

- **Gonadotrophins** are hormones that affect the gonads, the ovaries and testes. **Follicle stimulating hormone (FSH)** in the ovary of the female stimulates development of the follicles that contain eggs. In the male FSH stimulates the production and maturation of sperm in the testes.
- **Luteinising hormone (LH)** works with FSH in the female to bring about ovulation and to form a structure called the corpus luteum after ovulation. In the male LH stimulates interstitial cells in the testes to secrete male sex hormones.
- **Growth hormone (GH)**, or somatotrophin, stimulates body growth, particularly growth of the skeleton. It increases the rate at which amino acids are taken up by cells and built up into proteins. Growth hormone is secreted throughout life as it helps to maintain the size of organs once maturity is attained.
- **Thyroid stimulating hormone (TSH)**, or thyrotrophin, stimulates production and release of hormones from the thyroid gland.
- **Adrenocorticotrophic hormone (ACTH)**, or adrenocorticotrophin, controls production and release of some of the hormones from the cortex of the adrenal glands.
- **Prolactin**, or lactogenic hormone, works with other hormones to initiate and maintain milk secretion in females.

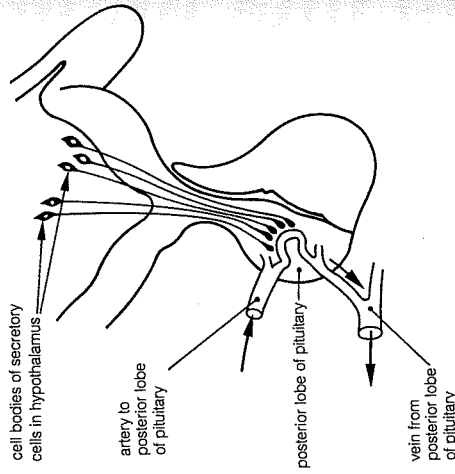


Fig. 24.3 Transport of hormones from the hypothalamus to the posterior lobe of the pituitary

Posterior lobe of the pituitary

The posterior lobe of the pituitary (neurohypophysis) releases the hormone oxytocin and antidiuretic hormone but neither are manufactured in the posterior lobe. Both hormones are produced in special cells in the hypothalamus. These cells have long axons that extend through the infundibulum to the posterior lobe. Hormones manufactured in the cell bodies move down the axons and are stored in the axon terminals ready for release into the bloodstream. Release of the hormones is triggered by nerve impulses initiated in the hypothalamus and conducted along the axons to the axon terminals (Fig. 24.3).

Oxytocin is a hormone that stimulates contraction of the muscles of the uterus. It is released in large quantities during labour. Oxytocin also stimulates contraction of cells in the mammary glands, resulting in release of milk during suckling.

Antidiuretic hormone (ADH), or vasopressin, causes the kidneys to remove water from urine that is forming. This water is returned to the bloodstream, so that ADH helps to retain fluid within the body. At higher concentrations ADH can also cause constriction of small arteries, the arterioles.

The thyroid gland

The thyroid gland is located in the neck just below the larynx. It consists of two lobes, which lie on either side of the trachea and are joined by a narrow piece of tissue that lies across the front of the trachea (Fig. 24.4).

The main hormone secreted by this gland is thyroxine. It is made from iodine and an amino acid and is continuously manufactured in the thyroid gland. Thyroxine controls body metabolism by regulating reactions in which complex molecules are broken down to release energy and in which complex molecules are synthesised from simple ones. The overall effect of thyroxine is to bring about release of energy and, since some of the energy released is in the form of heat, to increase body temperature.

The parathyroid glands

There are usually four parathyroid glands. Each is about the size of a small pea and they are

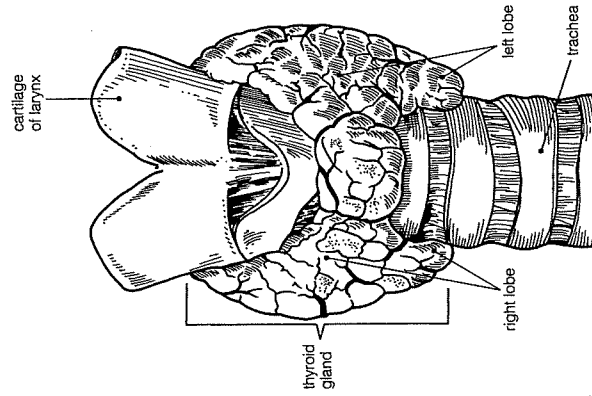


Fig. 24.4 Location of the thyroid gland

Fig. 24.5 Location of the parathyroid glands

embedded in the rear surface of the lobes of the thyroid gland (Fig. 24.5).

The parathyroid glands secrete **parathyroid hormone (PTH or parathormone)**, which controls calcium and phosphate levels in the blood.

The adrenal glands

There are two adrenal glands, one immediately above each kidney, as shown in Figure 24.6. Each adrenal gland has an inner **adrenal medulla** and an outer **adrenal cortex**. The two parts are different in structure and their functions are quite different. Thus, the adrenal gland is really two separate endocrine glands.

Adrenal cortex

Several different hormones are produced in the adrenal cortex but the two main ones are:

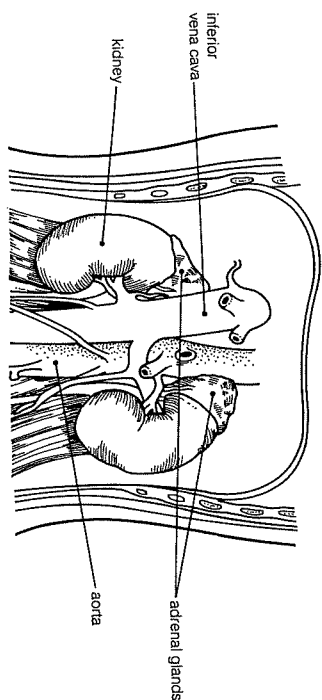


Fig. 24.6 Location of the adrenal glands

- **aldosterone**, which acts on the kidney to reduce the amount of sodium and increase the amount of potassium in the urine.
- **cortisol**, which, with related hormones, promotes normal metabolism and resistance of the body to stress

Adrenal medulla

Hormones produced by the adrenal medulla are adrenalin and noradrenalin.

- **Adrenalin** (epinephrine) has an effect similar to that of the sympathetic division of the autonomic nervous system. Adrenalin reinforces the activity of the sympathetic division in preparing the body for reaction to a threatening situation; that is, it is concerned with fight-or-flight responses.
- **Noradrenalin** (norepinephrine) has effects similar to those of adrenalin. In particular, it increases the rate and force of the heart beat.

The pancreas

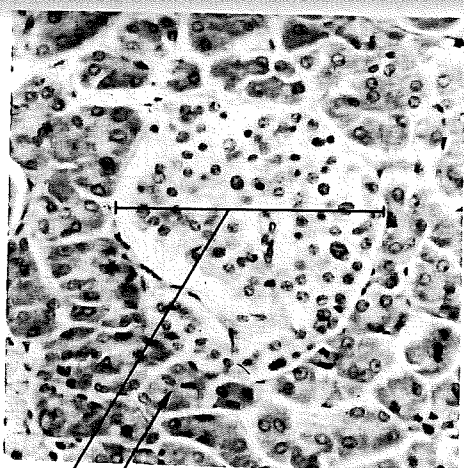
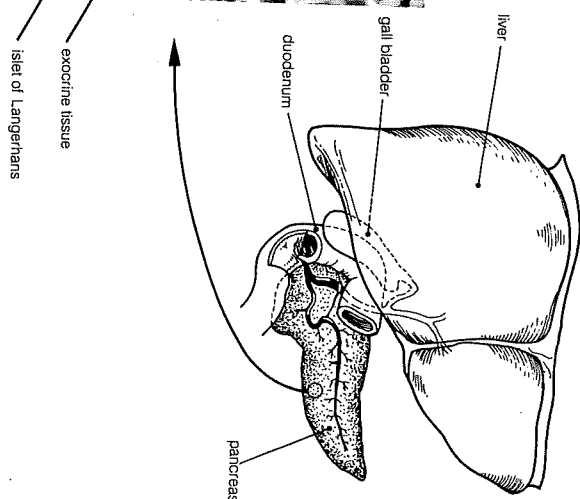
The pancreas lies just below the stomach and alongside the duodenum, the first part of the small intestine (Fig. 24.7). It is both an exocrine gland and an endocrine gland. The exocrine part secretes digestive enzymes into the small intestine through the pancreatic duct. Within the pancreas are clusters of special cells called **islets of Langerhans**. The islets are the endocrine part of the pancreas and they secrete two important hormones.

- **Insulin** has the effect of reducing the amount of glucose in the blood, that is, the blood sugar level. The reduction in blood sugar is achieved by promoting the uptake of glucose from the blood by the cells of the body. In the liver, insulin causes the conversion of glucose to glycogen and fat; in the skeletal muscles it causes formation of glycogen from glucose; in fat storage tissue it causes glucose to be converted into fat. The level of secretion of insulin from the pancreas is determined by the amount of sugar in the blood and is controlled through a negative feedback system.
- **Glucagon** acts in the opposite way to insulin. It works to increase blood sugar level, mainly by promoting the breakdown of glycogen to glucose in the liver. Glucagon also stimulates breakdown of fat in the liver and fat storage tissues.

The gonads

The gonads, testes and ovaries, produce hormones as well as sperms and eggs.

- **Androgens** are the male sex hormones produced by the testes. They are responsible for the development and maintenance of the male sex characteristics.
- **Oestrogens and progesterone** are the female sex hormones produced by the ovaries. They stimulate the development and maintenance of the female sexual characteristics. With the gonadotrophic hormones of the pituitary they also regulate the menstrual cycle and are

Fig. 24.7 Location of the pancreas, with photomicrograph of an islet of Langerhans
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involved in the changes that occur during pregnancy.

Other endocrine tissues

In addition to the major endocrine glands discussed above, there are other tissues, many of which are not called endocrine glands, that secrete hormones. For example, the stomach secretes the hormone gastrin, which stimulates the exocrine glands of the stomach to secrete gastric juice. The small intestine and kidney also secrete hormones. During pregnancy, the placenta secretes a number of hormones that help to maintain the pregnancy.

The pineal body in the brain and the thymus in the upper part of the chest are also believed to secrete hormones but more research is needed to determine their exact role.

Control of metabolic rate—an example of hormonal regulation

Metabolism is all the chemical reactions of the body—those that involve breakdown of complex molecules to *release energy* and those that *require energy* to build up complex molecules from simple ones. The **basal metabolic rate** is the total of all the metabolic processes taking place when a person is at rest. Thyroxine is the hormone directly involved with the regulation of basal metabolic rate. In particular, thyroxine increases the rate at which carbohydrates are broken down to release energy. Secretion of thyroxine is controlled by thyroid stimulating hormone (TSH) from the pituitary. Secretion of TSH is, in turn, controlled by factors from the hypothalamus.

What would happen if the body's basal metabolic rate decreased, or if the body needed more energy because the external environment

was cooler, or if there was a reduced level of thyroxine in the blood? The response to such a situation can be summarised as a series of steps.

1. In the hypothalamus, receptors detect a change in the chemical composition of the blood flowing through the brain.
 2. The hypothalamus secretes TSH releasing factor which is carried by blood capillaries directly to the anterior lobe of the pituitary gland.
 3. In response to the releasing factor from the hypothalamus the anterior lobe of the pituitary increases secretion of thyroid stimulating hormone.
 4. TSH is carried in the blood to all parts of the body but its target organ is the thyroid gland. The thyroid responds by secreting more thyroxine into the blood.
 5. The increased level of thyroxine in the blood causes an increase in metabolic rate and thus increased release of energy.
 6. The ultimate response, increased energy release, changes the original stimulus, which was an increased energy requirement. Thus, negative feedback has been achieved; the hypothalamus will detect the change in the blood and will act to reduce the secretion of thyroxine.
- The mechanism described maintains body processes in a steady state. Negative feedback ensures that metabolic rate can only fluctuate within very narrow limits. Figure 24.8 is a summary of this mechanism.

★ Don't need to know in detail

Control of the menstrual and ovarian cycles—cyclic control by hormones

As we saw in the previous section, feedback ensures that basal metabolic rate is maintained at a constant level. This was a steady state control system. In mature females changes occur in the ovaries and the uterus in a cycle. The cycle of changes that occurs in the uterus is called the **menstrual cycle** and it is integrated with changes that occur in the ovaries, the **ovarian cycle**. The cycles are typically twenty-eight days in length

but can be shorter or longer. These rhythmic changes are also controlled by hormones, but because there is a delay in the response to feedback, conditions in the ovary and uterus are not kept constant but change in a regular cycle. The sequence of events, and the hormonal changes that bring them about, are as follows.

1. The hypothalamus exercises ultimate control over both the menstrual and ovarian cycles through the secretion of gonadotrophin releasing factors. These factors are carried by the blood to the anterior lobe of the pituitary.
2. Gonadotrophin releasing factors stimulate the release of follicle stimulating hormone (FSH) and luteinising hormone (LH) from the anterior pituitary.
3. FSH stimulates the growth of a number of follicles in one of the ovaries. One of these will finally mature and release an egg.
4. Under the influence of FSH the follicle in the ovary secretes increasing amounts of oestrogens.
5. The higher level of oestrogens circulating in the bloodstream promotes growth of the lining of the uterus.
6. Oestrogen level in the blood becomes so high that secretion of FSH by the anterior pituitary is inhibited, while LH secretion is increased.
7. The change in the levels of FSH and LH result in **ovulation**—the release of the egg from the mature follicle.
8. The follicle from which the egg was released becomes a temporary endocrine gland called the **corpus luteum**. It secretes increasing quantities of oestrogens and progesterone.
9. The corpus luteum depends on secretions from the fertilised egg for its support. If fertilisation does not take place, the corpus luteum degenerates after about twelve days.
10. Secretion of progesterone and oestrogens drops because the corpus luteum is no longer functioning and this triggers off menstruation, the start of a new cycle. **Menstruation** is the breakdown of the lining of the uterus with a consequent loss of blood and tissue fluid. While menstruation is occurring, the ovarian cycle is also beginning again with the development of a follicle in one of the ovaries.

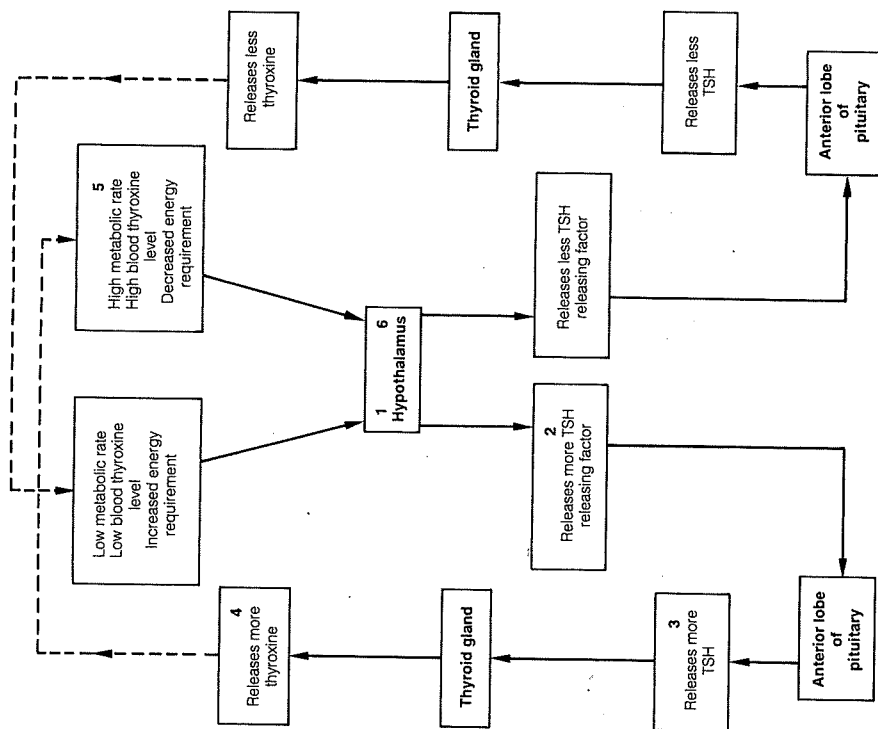


Fig. 24.8 Control of metabolic rate (the numbers refer to the steps outlined in the text)

Differences between hormonal and nervous co-ordination

In general terms it may be said that there are three differences between the actions of nerves and hormones in co-ordinating body activities.

- Nervous responses are more rapid than hormonal ones since nerve impulses travel rapidly along nerve fibres while hormones are transported in the bloodstream.
- Nerve impulses travel along a nerve fibre to a specific part of the body and often influence just one effector; hormones travel to all parts of the body, carried by the blood, and often affect a number of different organs.

On reaching an effector organ, nerve impulses bring about an immediate response, which lasts for only a short time; hormones are typically slower acting and responses can last a considerable time, even years. It must be stressed that these differences are only generalisations. There are exceptions to each of them. Reading through this chapter carefully you will be able to find some of the exceptions.

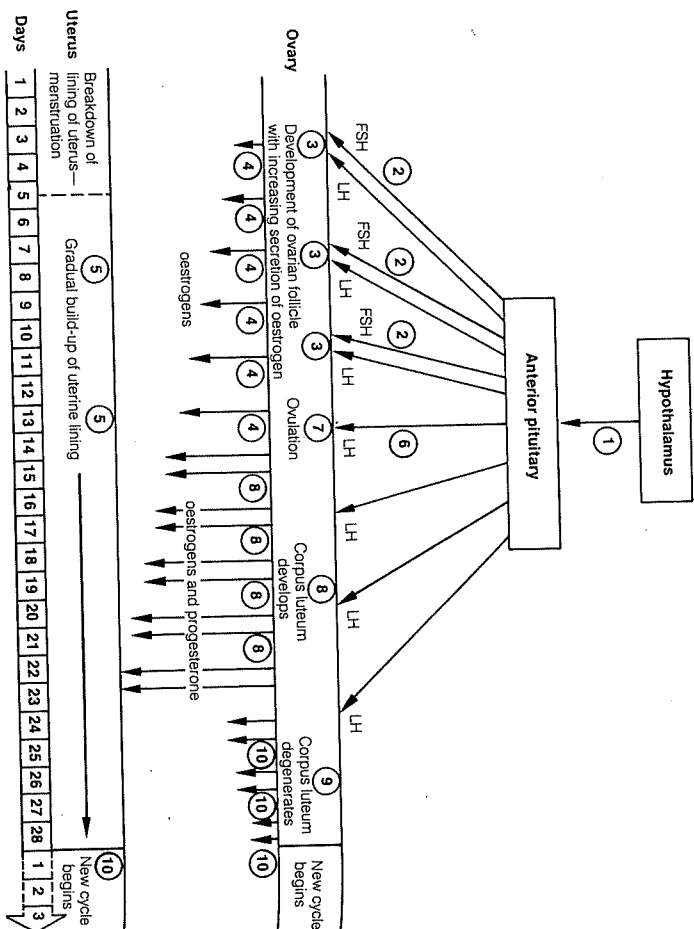


Fig. 24.9 Hormonal control of the ovarian and menstrual cycles

For example, response to the hormone adrenalin can be quite rapid; some hormones are not carried by the blood because their site of action is adjacent to the cells in which they are produced; some neurotransmitters, which carry impulses from motor nerve endings to effector organs, also act as hormones (e.g. adrenalin and noradrenalin).

Some disorders associated with endocrine glands

Most endocrine disorders occur because too much or too little of a hormone is produced. A few examples are discussed here; others will be mentioned in later chapters dealing more specifically with homeostasis.

- **Gigantism** results from oversecretion of growth hormone from the anterior lobe of the pituitary. The bones grow unusually long and the affected

person is abnormally tall. Undersecretion of growth hormone causes **dwarfism**, in which bones and other tissues fail to grow at the normal rate and affected persons are abnormally short.

- **Cretinism** is caused by undersecretion of thyroxine during the growing years. The brain and skeleton do not grow and develop in the normal way and so the person has retarded mental and physical development.
- **Simple goitre** is an enlargement of the thyroid gland, seen as a swelling in the neck. Thyroxine contains iodine, and if there is insufficient iodine in the diet, the thyroid gland enlarges in an effort to secrete more thyroxine.
- **Addison's disease** is caused by undersecretion of the adrenal cortex. Symptoms include muscular weakness, weight loss, mental lethargy and apathy.

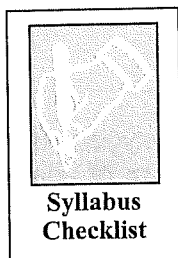
Review questions

1. Explain the difference between exocrine and endocrine glands and give five examples of each.
2. How do hormones change the activity of cells?
3. (a) What is a target organ?
(b) How do hormones get from their source to the target organ?
(c) What is the target organ, or organs, for each of the following hormones: luteinising hormone, prolactin, oxytocin, antidiuretic hormone, aldosterone, insulin and progesterone?
4. (a) What is negative feedback?
(b) Describe an example of negative feedback from those given in this chapter.
5. The hypothalamus controls release of hormones from the two lobes of the pituitary gland. Explain how control of the anterior lobe differs from control of the posterior lobe.
6. The pituitary is sometimes called the master gland because it secretes hormones that regulate the activity of other endocrine glands. Describe the pituitary hormones involved in the control of other endocrine glands.
7. Describe the role of the following hormones:
(a) oxytocin
(b) antidiuretic hormone
(c) adrenalin
(d) parathyroid hormone
(e) insulin
(f) glucagon
8. What are the male and female sex hormones and what are their roles?
9. Explain how basal metabolic rate is maintained at a constant level.
10. Describe the roles of the following hormones in the ovarian and menstrual cycles:
(a) follicle stimulating hormone
(b) luteinising hormone
(c) oestrogens
(d) progesterone
11. (a) Describe some of the common differences between nervous and hormonal coordination of body activities.

Apply your knowledge

1. Figure 20.5 shows the components of a steady state control system. Does the system for control of metabolic rate, which is described in this chapter, have each of these components? Explain.
2. Explain why endocrine glands are sometimes called ductless glands or glands of internal secretion.
3. Many famous people have suffered from endocrine disorders.
(a) John F. Kennedy, president of the United States from 1960 until his assassination in 1963, suffered from Addison's disease. Consult library references to see if you can find out some of Kennedy's medical history. How was he able to be president of the United States despite such a serious illness?
(b) Napoleon Bonaparte is believed to have suffered from a disease of the hypothalamus that caused the pituitary gland to function abnormally. Since the anterior pituitary regulates the functioning of the gonads and the adrenal and thyroid glands, these organs were also affected. See if you can find out the symptoms of Napoleon's disorder.
(c) Akhenaton, the Egyptian pharaoh who lived 3500 years ago, is portrayed in statues made later in his life with feminine features—prominent breasts, hips wider than the shoulders and a large amount of fat on the buttocks and thighs. It is believed that Akhenaton may have been afflicted with a disorder of one of the endocrine glands. Which gland, or glands, could it have been and what hormones could have been involved?
4. Figure 24.9 is a diagrammatic representation of the ovarian and menstrual cycles. It assumes the woman does not become pregnant. Using Figure 24.9 as a guide, construct a diagram of your own to show the changes that would take place if an egg were fertilised during the cycle and the woman became pregnant.
5. Simple goitre is common in a number of places throughout the world. Use library references to find:

2 HOMEOSTASIS



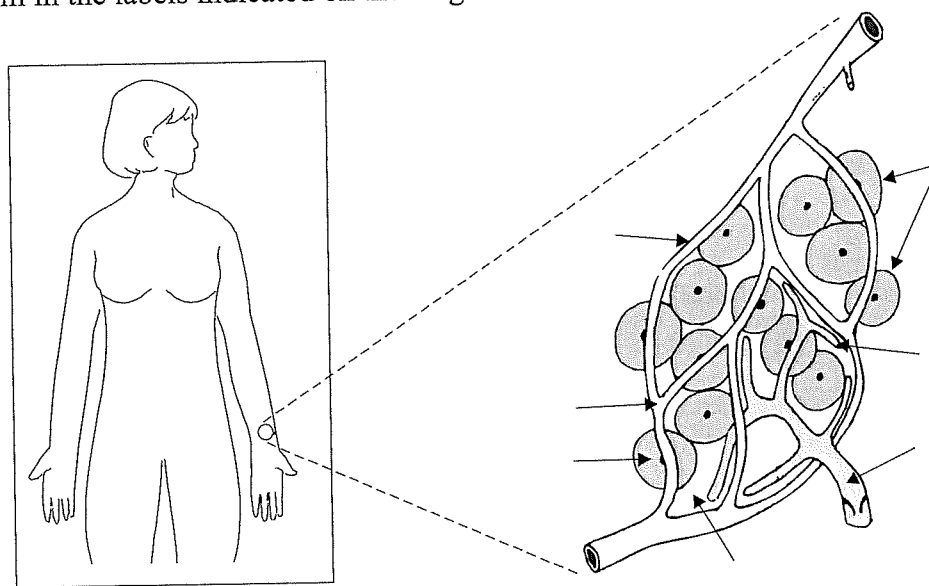
You should be able to:

- ✓ define homeostasis
- ✓ describe several differences between the internal and external environments
- ✓ explain the need for steady state control systems
- ✓ give examples of aspects of the body's internal environment that are maintained at reasonably constant levels through behavioural, physiological or cellular mechanisms
- ✓ describe the role of a steady state control system in the body
- ✓ identify the general components of steady state control system and describe their roles



REVIEW QUESTIONS

1. Fill in the labels indicated on the diagram below.

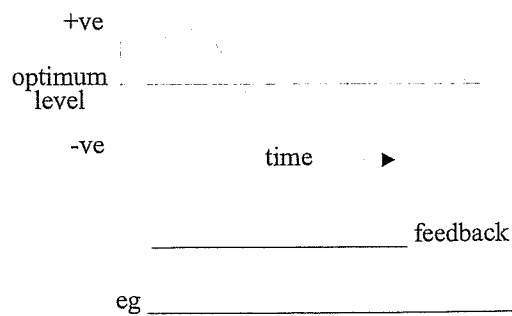
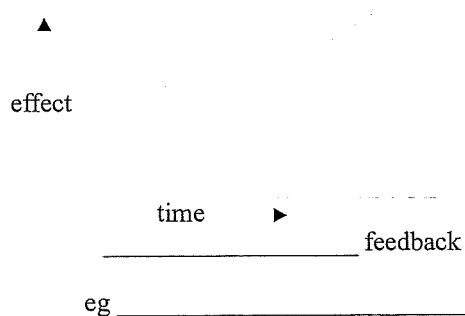


2. What substances can be found in the tissue (intercellular/interstitial) fluid?

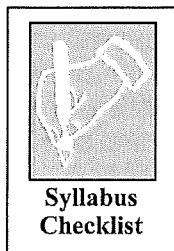
3. Define HOMEOSTASIS _____

OPTIMUM

4. The two body systems that regulate homeostasis are the _____ and _____.
5. Why is homeostasis so important?
- (a) _____
- (b) _____
- (c) _____
6. List some aspects of the body's internal environment that are kept at relatively constant levels.
- _____
- _____
- _____
7. What does a steady state control (feedback) system do?
- _____
- _____
- _____
8. List and describe the parts of a steady state system (feedback loop).
- _____
- _____
- _____
- _____
- _____
9. (a) Distinguish between positive feedback and negative feedback.
- POSITIVE FEEDBACK _____
- NEGATIVE FEEDBACK _____
- (b) Which of the two feedback systems is the most common in the body?
- _____
10. Label the graphs below indicating what type of feedback is represented.



5 ENDOCRINES



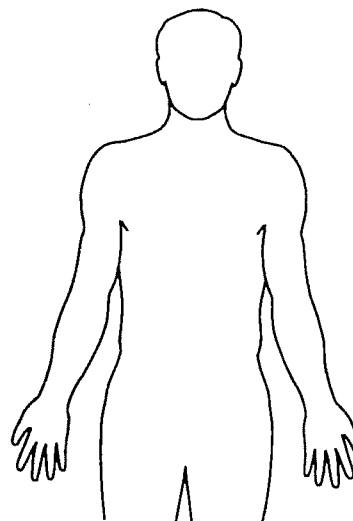
You should be able to:

- ✓ distinguish between exocrine glands and endocrine glands
- ✓ identify and locate the main endocrine glands in the body
- ✓ list the hormones produced from each endocrine gland and describe the function of each of the hormones they produce
- ✓ explain the hormonal control of :
 - the menstrual and ovarian cycles (cyclic control),
 - the basal metabolic rate of the body (steady state control) to demonstrate feedback loops
- ✓ describe differences between hormones and nerves with respect to the speed of transmission, specificity and persistence of action of both
- ✓ describe the relationships between the hypothalamus and pituitary, and the pituitary and the other endocrine glands.



REVIEW QUESTIONS

1. In the outline of a human figure, mark in and label the seven main endocrine glands.



2. Define the following:

GLAND _____

EXOCRINE GLAND _____

ENDOCRINE GLAND _____

HORMONE _____

TARGET CELL (OR ORGAN) _____

RECEPTOR SITES _____

ENDOCRINE GLAND	HORMONE	FUNCTION
Anterior pituitary anterior lobe	growth hormone	stimulates general body growth
	follicle stimulating hormone	stimulates maturation of ova in follicles, secretion of oestrogen by ovaries and production of sperm in testes
	luteinising hormone	ovulation in females, formation of corpus luteum and secretion of testosterone in males
	thyroid stimulating hormone	stimulates the thyroid to produce and release thyroxine
	adrenocorticotrophic hormone	stimulates the adrenal cortex to secrete hormones, mainly cortisol
	prolactin	breast development and milk production in females
Posterior pituitary posterior lobe	antidiuretic hormone	reabsorbs water back into blood from nephron
	oxytocin	causes contraction of smooth muscles of uterus during childbirth and release of milk from mammary glands
Parathyroid gland	parathormone (parathyroid hormone)	affects bone causing an increase in calcium levels in blood
Thyroid gland	thyroxine	stimulates metabolism
	calcitonin	lowers levels of calcium and phosphates in the blood by increasing uptake by bones
Pancreas (beta cells in islets of Langerhans)	insulin	decreases blood sugar level by increasing sugar uptake by cells, changes sugar into glycogen
(alpha cells in islets of Langerhans)	glucagon	increases blood sugar level, decreases sugar uptake by cells, changes glycogen into sugar
Adrenal cortex	aldosterone	increases reabsorption of sodium ions in kidney and increases excretion of potassium and hydrogen ions in urine (raises pH of blood)
	cortisol	helps to regulate metabolism and increases resistance to stress
Adrenal medulla	adrenalin (epinephrine) and noradrenalin (norepinephrine)	fight or flight response ie increases heart rate, blood pressure and respiration rate etc.
Ovaries	oestrogen	develop and maintain female sexual characteristics, help to regulate menstrual cycle and are involved in changes due to pregnancy
	progesterone	
Testes	testosterone	develop and maintain male sexual characteristics, sperm production

13. (a) Define METABOLISM. _____

- (b) What does the term METABOLIC RATE mean? _____

- (c) List some factors that affect metabolic rate. _____

- (d) Explain why an increased metabolic rate results in an increased body temperature.

14. Draw two feedback loops to show how the level of glucose in the blood is maintained in the body by insulin and glucagon:
- (a) when the glucose level is high
- (b) when the glucose level is low

15. Fill in the table below listing ways in which our bodies gain and lose heat.

MECHANISMS TO PRODUCE OR RETAIN HEAT	MECHANISMS TO LOSE HEAT

16. (a) As we exercise we often sweat profusely. How does sweating aid in cooling the body?

- (b) Why is sweating ineffective on humid days but quite effective on dry days?

17. Describe each of the following and explain how they contribute to heat loss from the body.

(a) radiation _____

(b) conduction _____

(c) convection _____



Trial Test 3: The Endocrine System

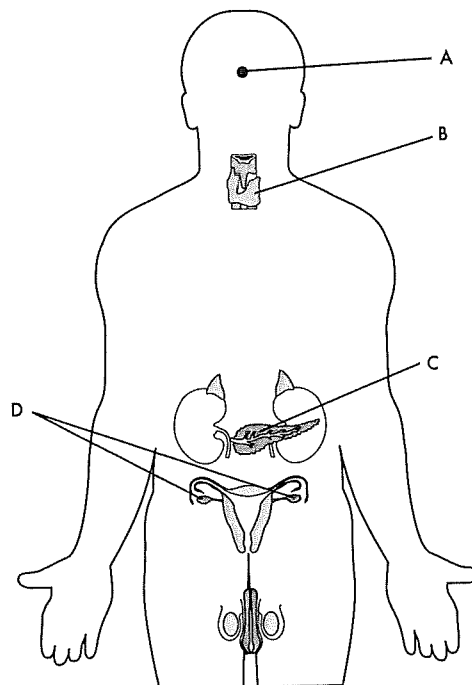
Time allowed: 50 minutes
Total marks: 100

Section 1 – Multiple Choice 30 marks
Section 2 – Short Answer 50 marks
Section 3 – Extended Answer 20 marks

Section 1 - Multiple Choice (30 marks)

1. An endocrine gland releases its products:
 - (a) directly into the blood.
 - (b) directly into the target organ.
 - (c) all the time.
 - (d) only at puberty

2. The products of endocrine glands carry out which of the following?
 - (i) Affect every cell in the body.
 - (ii) Affect only some cells and tissues.
 - (iii) Are carried by the blood.
 - (iv) Travel down ducts.
 - (v) Are only released on stimulation by the hypothalamus.
 - (a) (i) and (iii)
 - (b) (i), (iii) and (v)
 - (c) (ii) and (iii)
 - (d) (ii) and (v)



3. On the diagram above,
 - (a) A is the pituitary gland, B is the thyroid gland, C is the pancreas and D is the testes.
 - (b) A is the pituitary gland, B is the parathyroid gland, C is the pancreas and D is the adrenal gland.
 - (c) A is the pituitary gland, B is the thyroid gland, C is the pancreas gland and D is the ovary.
 - (d) A is the pituitary gland, B is the parathyroid gland, c is the ovary and D is the testes.

- (a) insulin.
- (b) calcitonin.
- (c) thyroxine.
- (d) glucagon.

- adrenal medulla.
- adrenal cortex.
- anterior lobe of the pituitary gland.
- posterior lobe of the pituitary gland.

- (i) Generally have low blood sugar.
- (ii) Generally have high blood sugar.
- (iii) Can control their blood sugar levels with diet.
- (iv) Can control their blood sugar levels with insulin.
- (v) Can control their blood sugar levels with glucagon.

(a)	(i) and (iii)	(b)	(i) and (iv)
(c)	(ii) and (iii)	(d)	(ii) and (v)

Which of the flow diagrams below shows this information?

- (a) Hypothalamus → pituitary → adrenal cortex → cortisol
- (b) Hypothalamus → pituitary → adrenal cortex → cortisol
 ↑
 feedback
- (c) Hypothalamus → pituitary → adrenal cortex → cortisol
 ↑
 feedback
- (d) Hypothalamus → pituitary → adrenal cortex → cortisol
 ↑
 ↑
 feedback

- (a) Anterior pituitary → human growth hormone → body cells.
- (b) Anterior pituitary → melanin stimulating hormone → skin cells.
- (c) Anterior pituitary → follicle stimulating hormone → production of sperm in testes.
- (d) Anterior pituitary → antidiuretic hormone → kidney tubules.

Which statement about it is correct?

- (a) It is controlled by negative feedback.
- (b) It is controlled by positive feedback.
- (c) Oxytocin is produced from the anterior pituitary.
- (d) Oxytocin controls the release of milk from the breast.